Mini project by Python documentation

Mini-Project 1: Advanced Python Data Exploration & Automated Reporting

1. Project Overview

This project analyzes the **Sample Superstore 2019** dataset using Python. The main goal is to explore sales, profit, customers, discounts, returns, and business performance, then generate useful analytical insights and visualizations.

The project was implemented using:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

2. Dataset

The Excel file contains three sheets:

- **Orders:** Contains main transaction data (9,994 records, 21 columns including Order ID, Order Date, Ship Date, Customer, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit).
- **People:** Contains regional sales managers (4 records).
- **Returns:** Contains returned orders information (800 records).

```
 # Data Loading Code
orders = pd.read_excel('Sample - Superstore 2019.xls', sheet_name='Orders')
people = pd.read_excel('Sample - Superstore 2019.xls', sheet_name='People')
returns = pd.read_excel('Sample - Superstore 2019.xls', sheet_name='Returns')
```

3. Data Inspection

The dataset was inspected using Pandas functions: `head()`, `shape`, `info()`, `isnull().sum()`, and `duplicated().sum()`.

4. Data Cleaning

The following cleaning operations were performed:

- **Date Conversion:** Converted `Order Date` and `Ship Date` to datetime.
- **Missing Values:** Handled missing values in `Postal Code` using `fillna(0)`.
- **Duplicate Records:** Removed duplicate records using `drop_duplicates()`.

```
 # Data Cleaning Code
orders = orders.drop_duplicates()
orders["Order Date"] = pd.to_datetime(orders["Order Date"])
```

```
orders["Ship Date"] = pd.to_datetime(orders["Ship Date"])
orders["Postal Code"] = orders["Postal Code"].fillna(0)
```

5. Feature Engineering

New columns created:

- **Profit Margin:** `(Profit / Sales) * 100`
- **Shipping Duration:** `(Ship Date - Order Date).dt.days`
- **Sales Performance:** Categorized into *Low*, *Medium*, *High*, and *Very High* using `pd.cut()`.
- **Year:** Extracted using `Order Date.dt.year`.

```
 # Feature Engineering Code
orders["Profit Margin"] = (orders["Profit"] / orders["Sales"]) * 100
orders["Shipping Duration"] = (orders["Ship Date"] - orders["Order Date"]).dt.days
orders["Sales Performance"] = pd.cut(orders["Sales"], bins=[-np.inf, 50, 200, 500, np.inf], labe
orders["Year"] = orders["Order Date"].dt.year
```

6. KPI Analysis

KPI	Value
Total Sales	$2.30M ($2,297,200.86)
Total Profit	$286.40K ($286,397.02)
Total Quantity	37,873
Total Orders	5,009
Profit Margin	12.47%
Average Order Value	$458.61

7. Category Analysis

Technology generated the highest sales and profit, while **Furniture** had a significantly lower profit margin.

8. Region Analysis

West region generated the highest sales and profit margin, while **Central** had the lowest profit margin.

9. Customer Segment Analysis

Consumer generated highest total sales, but **Home Office** achieved the highest profit margin.

10. Discount Analysis

Discount Range	Profit Margin (%)
0–10%	28.89%
10–20%	11.58%

20–30%	-10.05%
30–50%	-24.80%
50%+	-119.20%

11. Time Analysis

Sales increased considerably from 2016 to 2019, showing steady enterprise growth.

12. Correlation Analysis

Performed across numerical fields (Sales, Profit, Quantity, Discount, Shipping Duration) to analyze inter-variable dependencies.

13. Returns Analysis

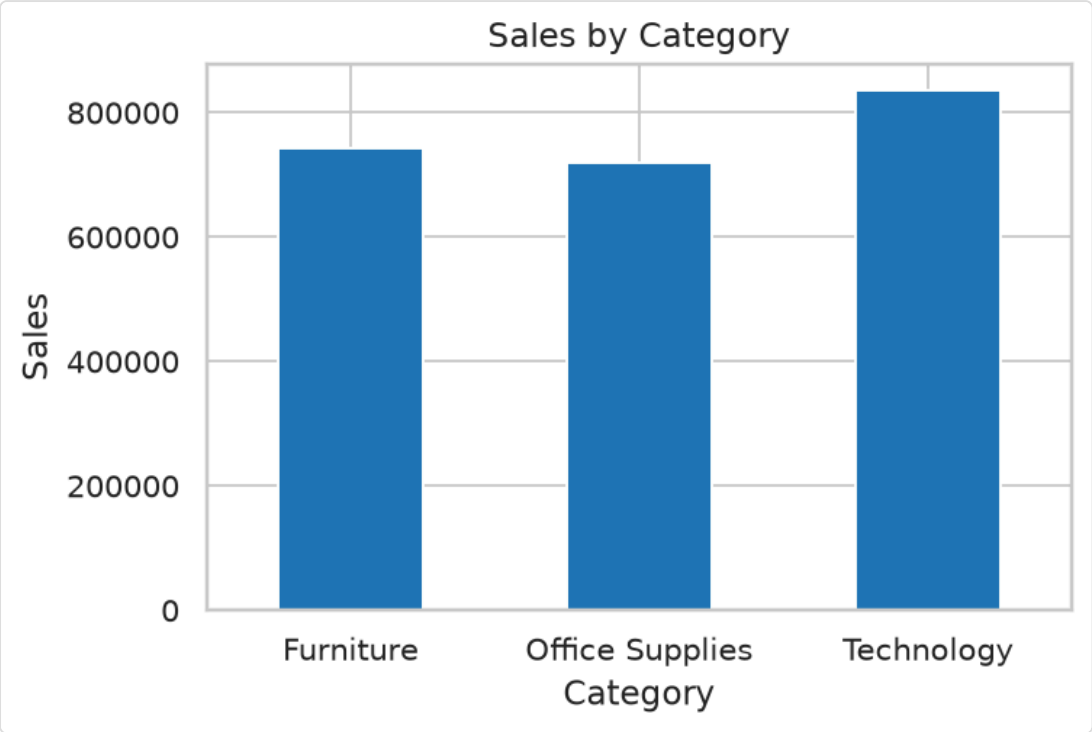
Joined Returns sheet with Orders. Total returned orders: **800**, with an overall return rate of approximately **8.00%**.

14. Outlier Analysis

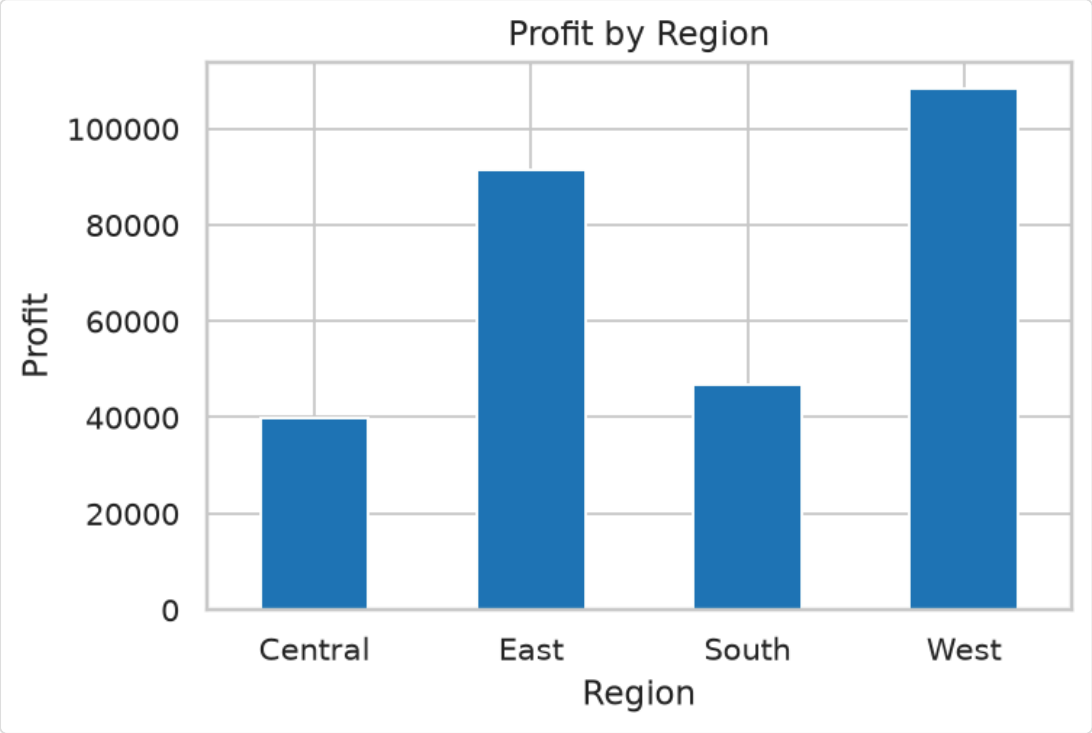
Using Interquartile Range (IQR): identified **1,167** sales outlier transactions.

15. Data Visualizations (11 Charts)

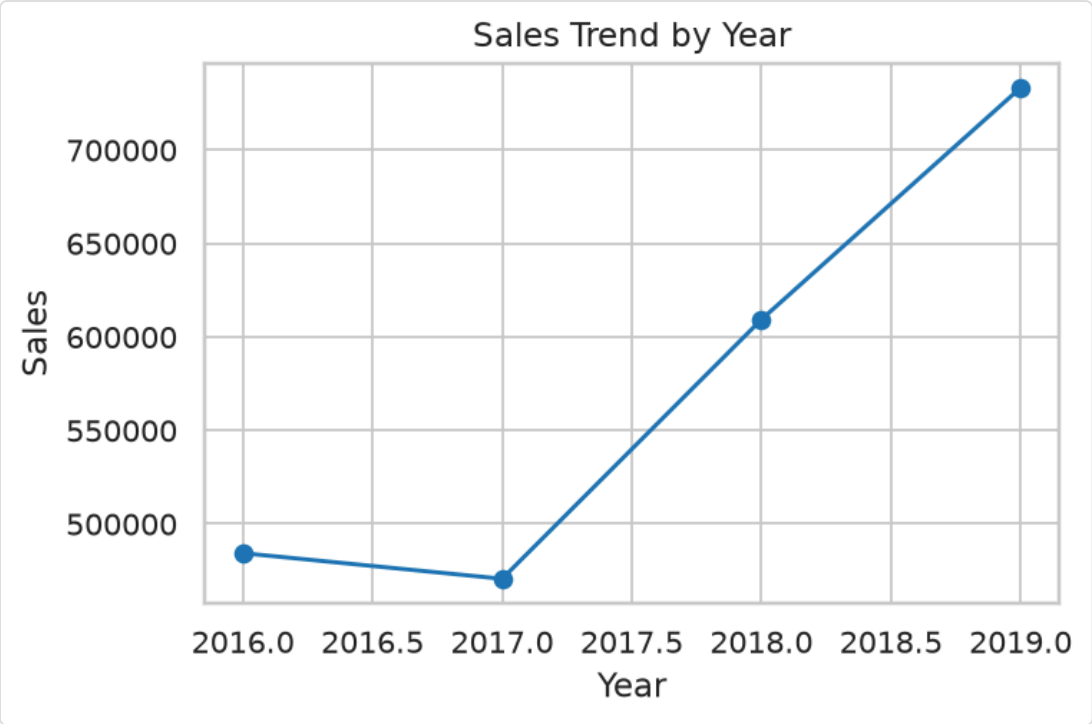
1. Sales by Category



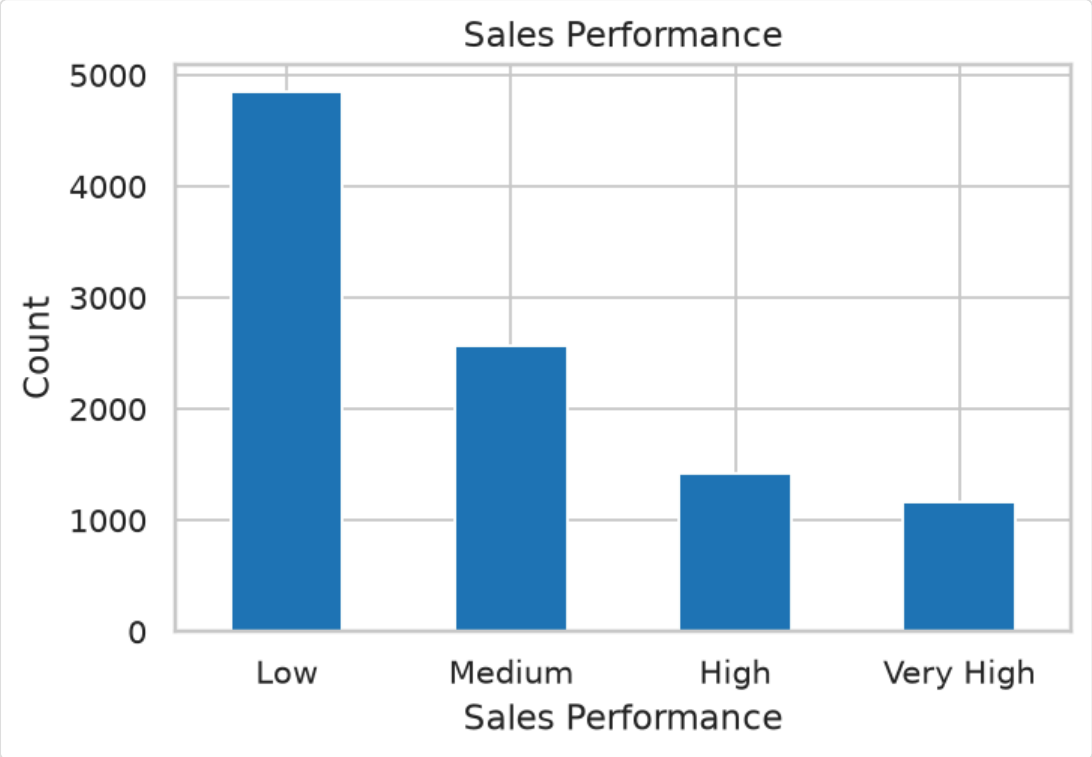
2. Profit by Region



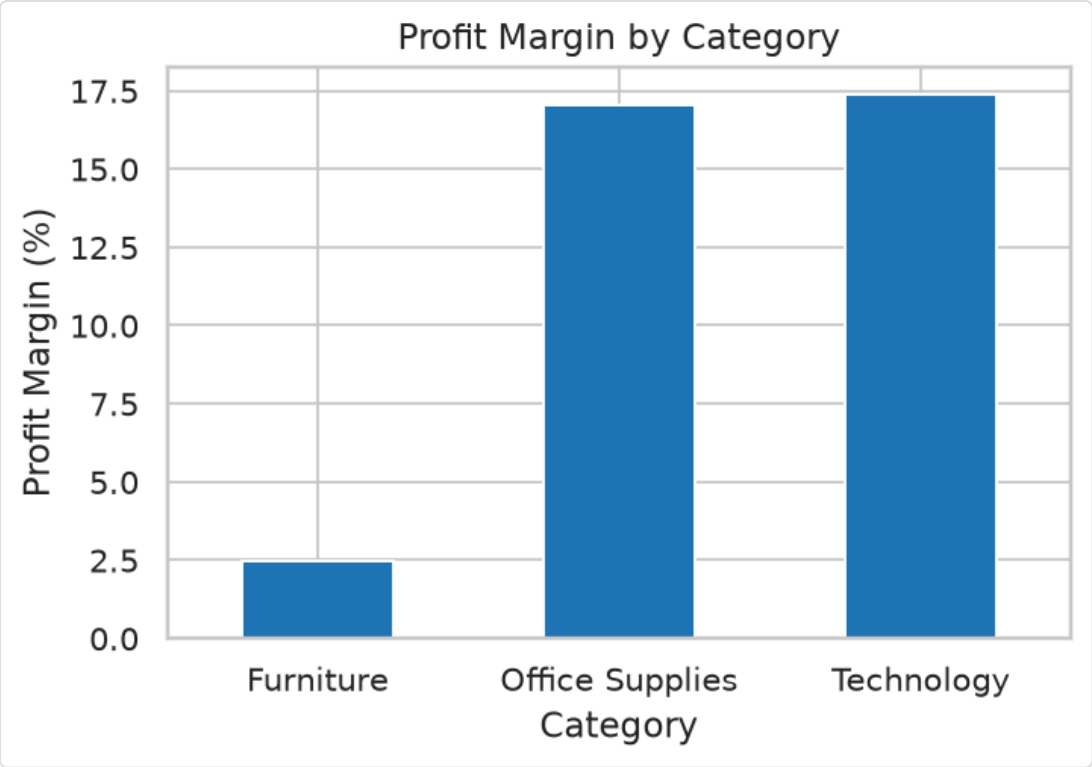
3. Sales Trend by Year



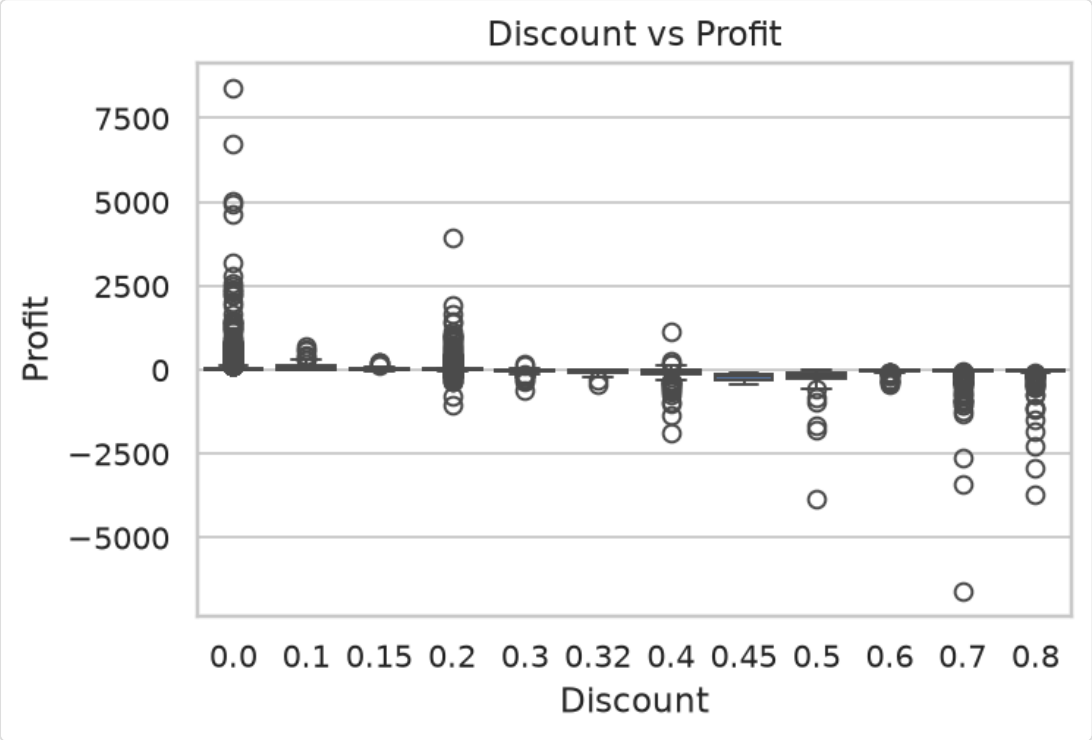
4. Sales Performance



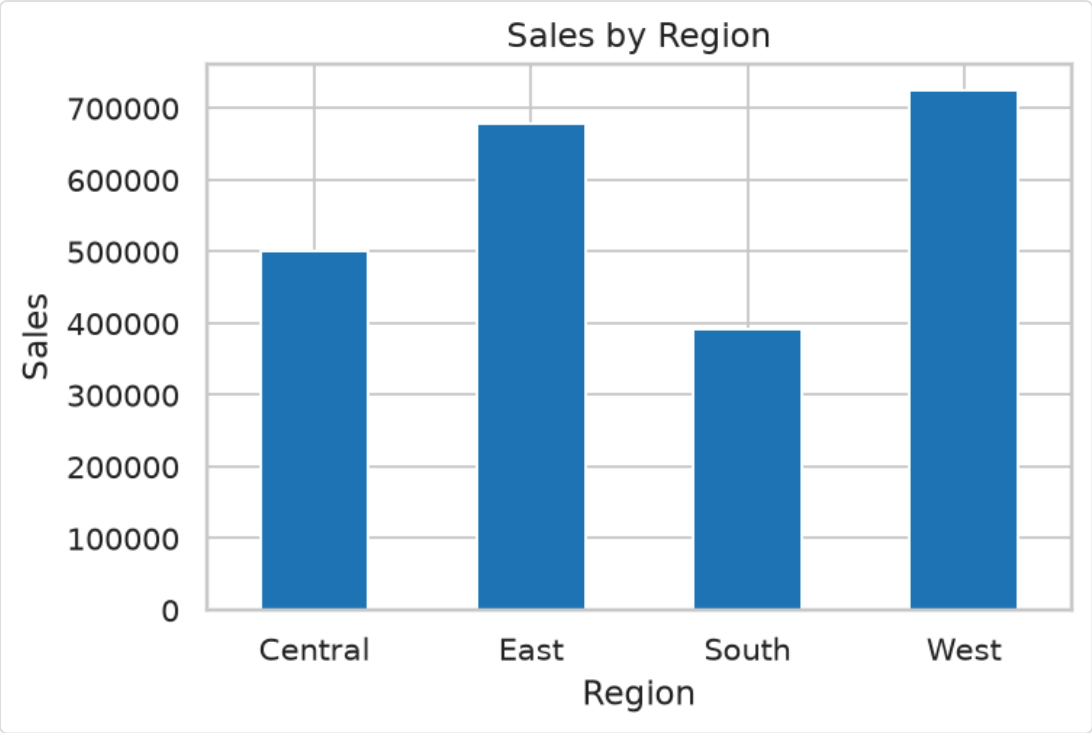
5. Profit Margin by Category



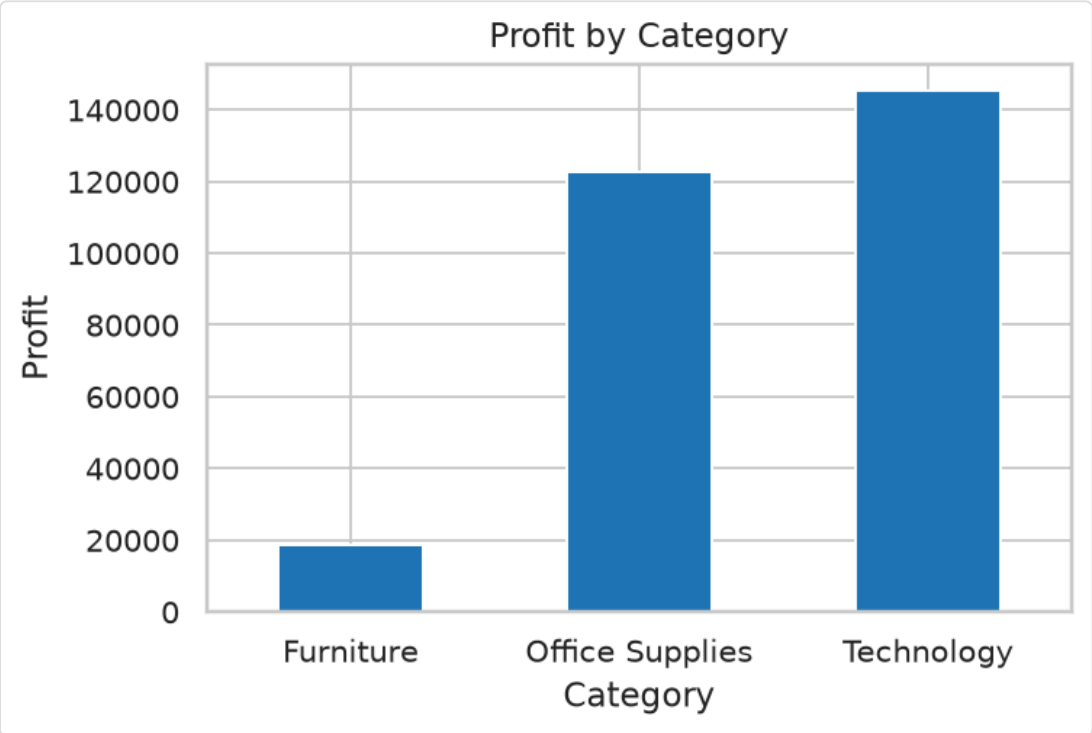
6. Discount vs Profit



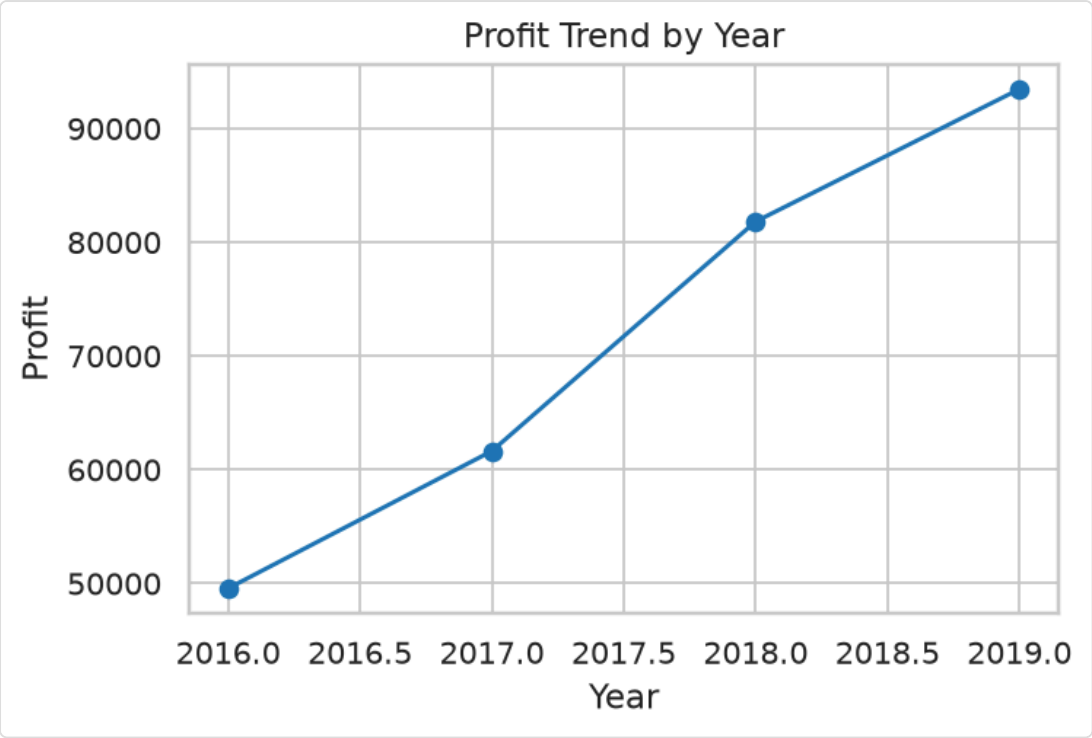
7. Sales by Region



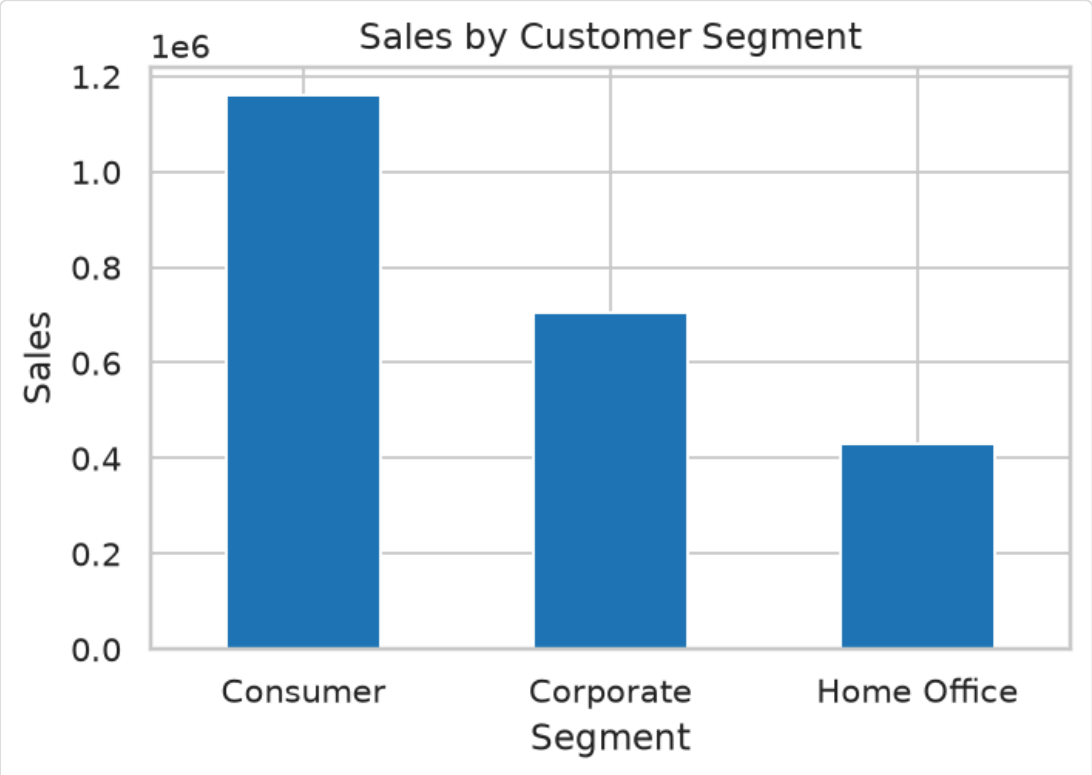
8. Profit by Category



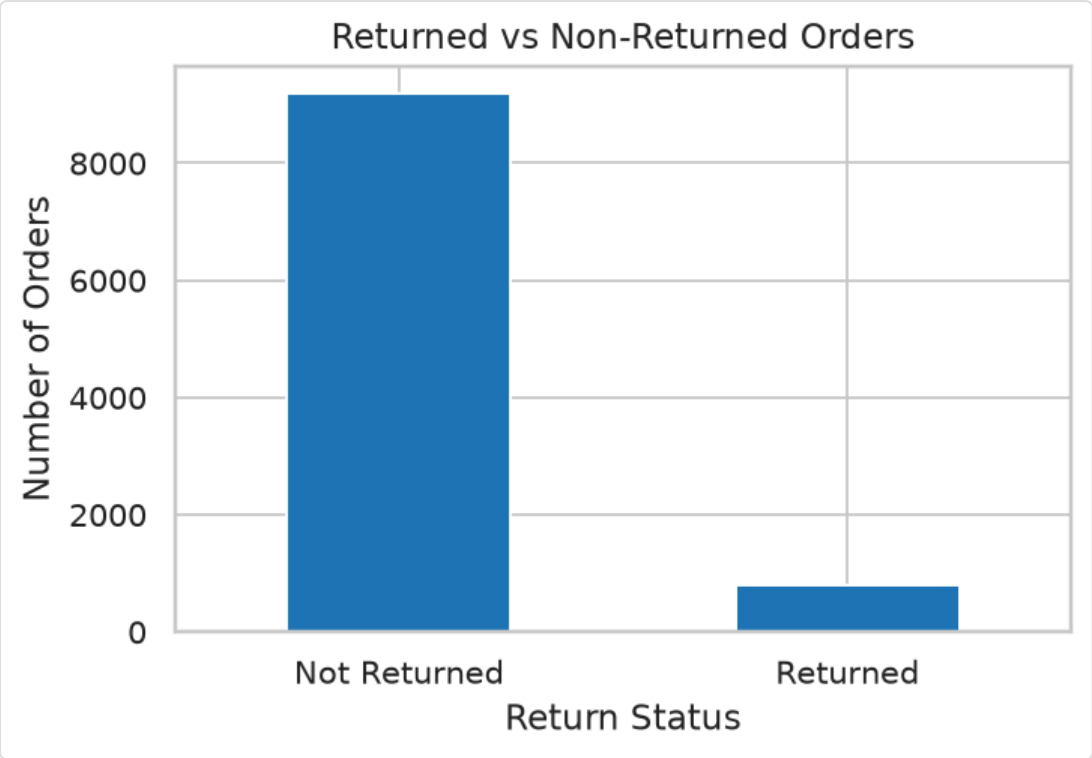
9. Profit Trend by Year



10. Sales by Customer Segment

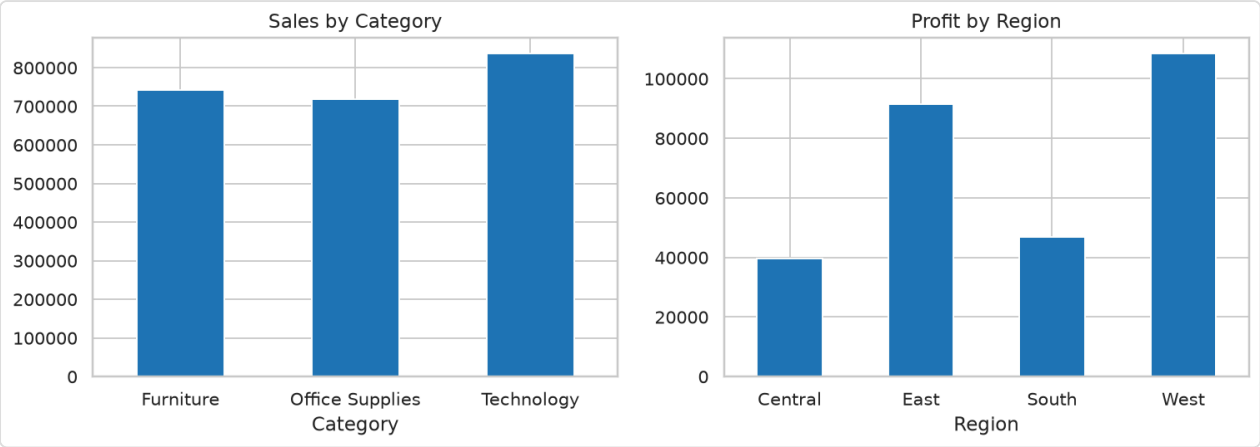


11. Returned vs Non-Returned Orders

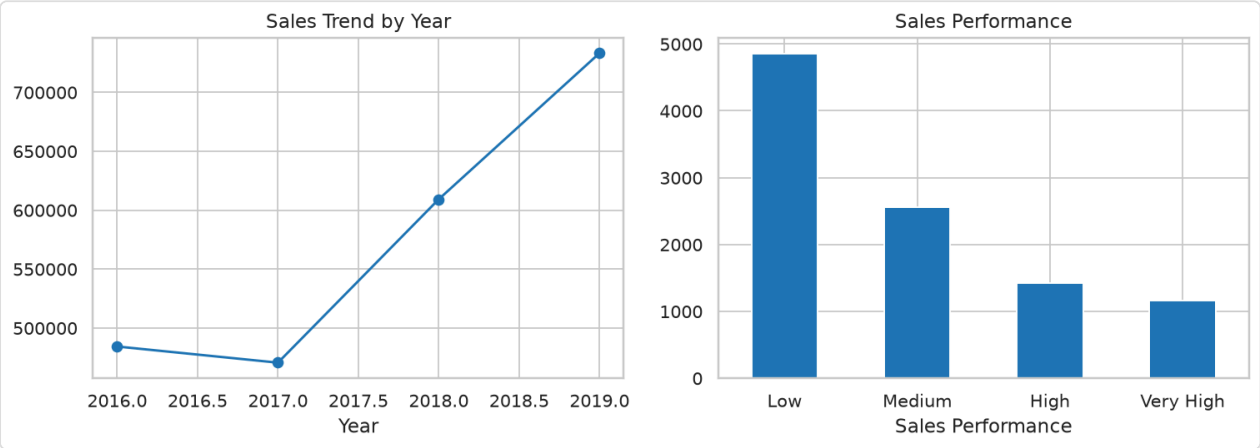


16. Dashboards

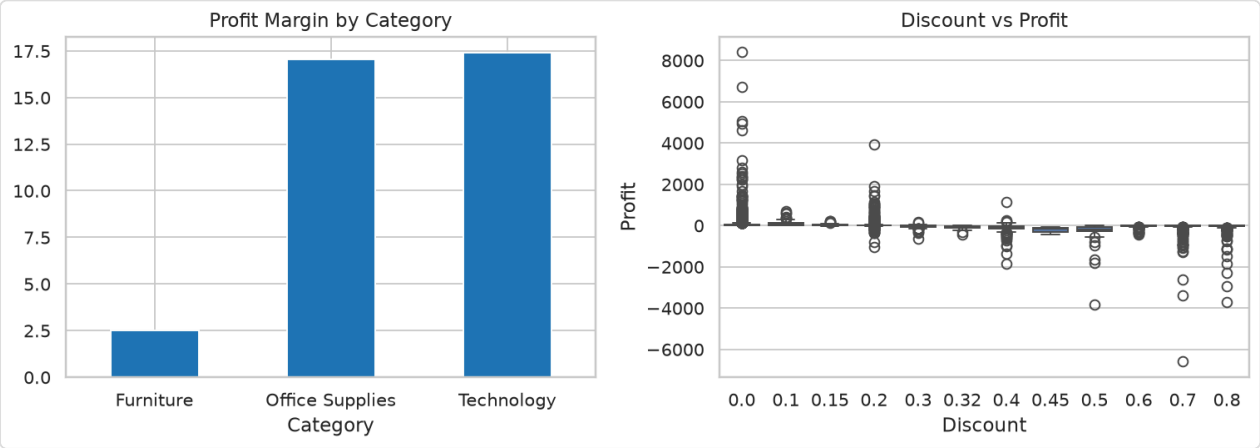
Dashboard 1 — Executive Overview (Sales by Category & Profit by Region)



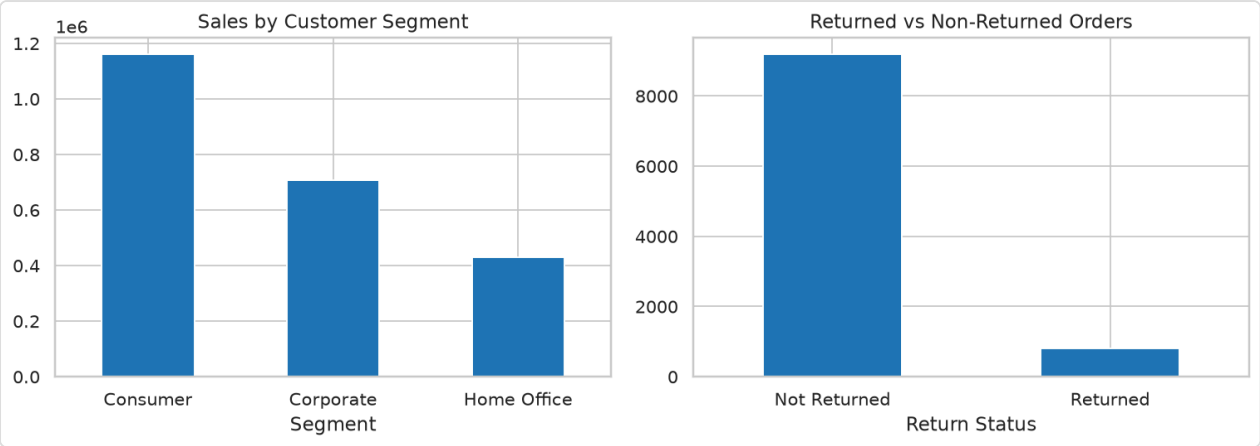
Dashboard 2 — Sales Performance (Sales Trend & Performance Category)



Dashboard 3 — Profitability (Profit Margin by Category & Discount vs Profit)



Dashboard 4 — Customers & Returns (Segment Sales & Returns Comparison)



17. Main Business Insights

- **Technology** is the strongest performing category in both sales and profitability.
- **Furniture** demonstrates low profitability margins requiring discount policy revision.
- **West** region generates maximum profit and revenue.
- **Discounts over 20%** consistently generate operational losses.
- Annual sales and profits exhibited consistent upward growth from 2016 through 2019.

18. Conclusion & 19. Technologies Used

`Data Import → Data Inspection → Data Cleaning → Feature Engineering → KPI Analysis → EDA → Visualizations → Dashboards → Business Insights`

Technology	Purpose
Python	Main programming language
Pandas	Data manipulation and transformation
NumPy	Numerical operations and vectorized computing
Matplotlib	Core chart rendering and figure structuring
Seaborn	Statistical distribution plots and heatmaps
Excel	Source dataset storage

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